

Laser-based Systems for Onboard Detection of Flange Grease on Revenue Service Tracks

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Abstract:

The results of an extensive study to detect the presence of flange-face grease on revenue service tracks are presented. This study aims to satisfy the rail industry's need to determine track lubrication conditions in a cost-effective and field-implementable manner. The system promises to provide alternatives to the current empirical and subjective methods in which the track engineer's periodic visual inspection is used to assess flange lubrication in curves. The mobile system uses two laser-based optical systems onboard a moving platform such as a track cart, Hyrail truck, or track geometry car to measure the presence or absence of fluorescence on the rail flange. The flange grease condition is determined according to the intensity of fluorescence detection on a tri-level scale, indicating low/no, medium, and high amount of flange lube. The successful test results indicate good correlation with independent tribometer measurements that are commonly used as the most accurate means of detecting flange lube. Whereas the tribometer measurements are slow, labor intensive, and limited to small track sections, the laser-based system developed in this study can be used at track speeds over extended sections of track. Although this study was focused on the system development and validation, the successful commercial implementation of this system can significantly advance railroad practices to detect dried curves before they lead to significant wheel flange and rail wear.

Introduction

Effective lubrication of rail surfaces, especially on curved tracks, is essential in the railway industry to reduce rail wear and maintain safe, efficient operations. In particular, flange-face lubrication is critical for minimizing the wear between the wheel flange and the rail's flange face, especially on tight curves, where high lateral forces develop between the wheel and rail [1,2]. Furthermore, flange lubrication also plays a key role in preventing wheel-climb derailments [1,3,4].

As the train negotiates a curve, the outer rail—known as the “high-rail”—is subjected to greater wear as the wheel flange pushes against the rail's inner head corner, under the influence of these lateral forces. To mitigate this wear and extend the life of wheels and rails while also lowering maintenance costs [5], flange lubrication is essential. In addition, it helps reduce rolling contact fatigue on the rail head [6] lowers wheel-squeal noise [7], and decreases fuel consumption [8,9]. While the inner rail, or “low-rail”, also experiences wear, it is primarily due to contact with the wheel tread rather than the flange. This wear results from creep forces, friction and load distribution as the train moves through the curve [10].

Figure 1. Detailed representation of rail surface geometry with key regions and angular gradients

Grease is applied to the flange-face of rails using wayside applicators [11]. These devices squirt grease onto the rail as a train approaches and the rotating wheels then spread it along the track. Applicators are strategically positioned before tight curves which are known to cause high wear, to reduce friction between the wheel flange and rail side. However, if an applicator malfunctions—due to issues like a broken pump or an empty grease reservoir—it may go unnoticed until significant wear becomes visible. An Australian railway study [12] revealed that around 25% of the grease applicators in the study area were not functioning properly.

Currently, there are limited methods for assessing flange-face lubrication levels. The most common approach is visual inspection for checking the presence or absence of grease. Although simple, this method is often inaccurate, time-consuming, and prone to human error. It also struggles to distinguish between fresh, effective grease and old, dried-up grease that has lost its fluid properties and no longer reduces friction. Another method of inspection is by using tribometers which offer quantitative measurements of rail friction. However, they can only be used over short sections of track at walking speeds and may intervene with train operations. Additionally, they do not measure grease thickness, which is vital for estimating how much grease remains available to maintain proper lubrication on the rail surface.

As an alternative to the current inspection methods, Railway Technologies Laboratory (RTL) located at Virginia Tech in Blacksburg, VA, has developed an optical sensing system to detect rail grease [13]. This system can be easily installed on Hyrail trucks or other track vehicles commonly used by maintenance crews and engineering staff, thus allowing for the assessment of rail lubrication over extended distances. It utilizes a technique

called the "laser-induced fluorescence optical method (LFOM)" to identify the presence of flange-face grease and measure its relative thickness. The details about the RTL system are presented in the next section.

Design and Functionality of RTL’s Laser-Based Grease Detection System

Industrial rail greases are typically manufactured with a hydrocarbon base stock, giving them low viscosity, which helps them spread effectively along the rail [14]. These specially designed greases often exhibit fluorescent characteristics, meaning they emit light when exposed to a specific wavelength [15]. This fluorescent response, detectable by a fluorescence sensor, stems from the grease's hydrocarbon-based chemistry and the liquid “carrier” compounds used in flange greases across North America and likely in other regions as well. Even greases that are not hydrocarbon-based can show fluorescence due to their organic chemical makeup.

In accordance with these chemical properties of the grease, an optical sensing system was developed by RTL that detects the fluorescence light emitted by grease on the flange face to determine its presence or absence on the rail. This system includes two 405 nm lasers, a fluorescence band-pass filter, and a fluorescence optical detector. The lasers are linearly polarized and arranged in a cross-polarized configuration to reduce laser speckle. Although the lasers operate separately, their wavelengths are nearly the same to ensure effective detection of rail grease fluorescence. Slight differences in the laser wavelengths help prevent speckle modulation, which refers to unwanted variations in intensity caused by interference when coherent laser light reflects off uneven surfaces. These variations can interfere with accurate fluorescence detection by introducing noise into the signal. These coherent beams are directed onto the flange face of the rail, creating an illuminated volume within the grease, defined by the grease thickness and the beam's spot size on the rail. The optical detector is positioned approximately at a 60-degree angle to the longitudinal rail surface, a placement found—through multiple beam configuration tests—to offer the highest signal-to-noise ratio (SNR) for detecting emitted fluorescence. More details on how the RTL system operates are available in [13].

Two lubricity detection devices, referred to as “Unit 1” and “Unit 2,” were fabricated to monitor the lubrication levels on the left and right rails, as illustrated in. Each unit uses the same type of laser emitters and filters and is positioned with the same geometric alignment and angle toward the rail surface, with the sole difference that one unit is the mirror image of the other in terms of beam’s direction along the track. This mirrored setup ensures that both rails are inspected at the same angle relative to the motion of the cart. Both units were mounted on a controlled robotic cart, as shown in Figure 3.

Figure 2. The inner assemblies of Unit 1 and Unit 2 are designed to project laser beams onto the side of the railhead, creating an illuminated spot with a diameter of 1.5 cm.



Figure 3. (a) RTL system mounted on a remote-controlled rail cart; (b) Schematic representation of the optical sensing unit for lubrication detection.

Mobile Rail Cart Configuration for RTL System

The remotely operated rail cart used for testing is equipped with an electric propulsion and braking system to ensure safe use on active rail tracks [16]. It can travel at speeds of up to 10 mph (16 km/h). A tachometer installed on the front axle tracks wheel rotations, allowing for precise calculations of both speed and distance traveled. This data is collected at a rate of 125 Hz, ensuring high-resolution distance measurements.

As shown in Figure 3a, the two detection units are mounted ahead of the front wheels—Unit 1 on the right rail and Unit 2 on the left—so that grease measurements are taken before the cart’s wheels and flanges can disturb the grease layer. In Figure 3b, each unit is angled at 66 degrees from the vertical and positioned at a standoff distance of 5 inches (12.7 cm). This arrangement directs the laser beams onto specific areas along the side of the railhead where flange-face grease is typically applied, as highlighted in Figure 3a.

Performance evaluation of RTL system through dynamic laboratory tests

We conducted a series of laboratory tests to assess the performance of the sensing system under simulated operating conditions. The tests were performed on a 40 ft track panel at the RTL facility. Both rails of the panel were applied with a standard grease sample commonly used by the US railroad industry. To evaluate the RTL system’s sensitivity to varying lubrication levels, the grease was manually applied in three distinct 3-foot sections with increasing quantities: lightly applied (thin coating), moderately applied (intermediate thickness), and heavily applied (thick coating). The sections were separated by a clean, ungreased segment of rail to clearly distinguish the fluorescence readings associated with each level of grease. This setup enabled controlled comparison of fluorescence signals across different grease volumes. The RTL system (Unit 1 and Unit 2) was securely mounted on the rail cart, as shown in Figure 3. The cart was then driven along the length of the test track panel, allowing the system to collect continuous fluorescence readings as it passed over the differently greased sections of the rail.

Figure 4. Track panel setup at RTL used for performing dynamic laboratory tests to evaluate RTL system’s sensitivity to varying lubrication levels.

The results of the dynamic laboratory test are illustrated in Figure 5. The plot displays the fluorescence intensity voltage recorded by Unit 1 (mounted on the right side). For brevity, only the results from Unit 1 are presented as both the units produced identical measurements. A clear distinction in fluorescence intensity voltage was observed across the lightly, moderately, and heavily lubricated sections. The recorded signal increased proportionally with the amount of grease applied, with the lightly lubricated section showing the lowest intensity, the moderately lubricated section showing intermediate values, and the heavily lubricated section producing the highest intensity. This trend demonstrates the RTL system’s sensitivity to varying levels of lubrication.

Figure 5. Fluorescence intensity voltage recorded by Unit 1 (mounted on the right side of rail cart) during the laboratory testing on the RTL track panel, showing clear differentiation between light, medium and heavy grease applications, separated by clean rail sections.

Low-speed field tests

We conducted a series of field tests at low-speeds, on active revenue service tracks to assess the system’s ability to detect lubrication under dynamic conditions outside the lab. The tests aimed at exposing our system to actual rail environments, including operational surfaces, tight curves and naturally occurring grease patterns. The following subsections provide details on the test setup, testing locations, and the methodology used during the field evaluations.

Test Setup

For the field tests, Unit 1 and Unit 2 were mounted on the rail cart, one on each side, to simultaneously assess grease conditions on both rails, as shown in Figure 6. The rail cart, designed for remote control, enabled lubricity detection tests to be conducted at up to 10 mph speeds. For low-speed testing the cart was typically driven at 2 to 5 mph speeds. A Global Navigation Satellite System (GNSS) receiver was mounted on the rear of the cart frame to track the rail cart’s path and speed, as shown Figure 17. A tachometer installed on the front axle of the cart was used to measure the rotation of the axle

to measure the instantaneous vehicle speed and travel distance more precisely than what is available from the GNSS. Cameras by GoPro, Inc. manufactured in San Mateo, California, USA, were also installed on the cart to visibly record the lubricity conditions of both the left and right rails.

Figure 6. RTL system mounted on rail cart to conduct low-speed field tests on revenue service tracks.

Field test section 1 (Christiansburg, VA)

We conducted the first round of low-speed field tests on a Class I railroad in collaboration with Norfolk Southern (NS) in Christiansburg, VA. Among the two available tracks (Track #1 and Track #2) at the facility, the test was conducted on Track #2 as shown in Figure 7. The figure illustrates the test route, cart setup and change in elevation of the path covered. During the test, the rail cart was initially run from Point A to Point B, covering a total distance of 1.07 miles and passing through a rail grease applicator, identified by the green dot in Figure 7. The cart was then turned around and operated from Point B to the starting point, Point A.

Figure 7. Track layout of test section 1 (Christiansburg, VA). Top left corner graph represents the speed of the cart along the distance travelled during the test; Bottom right image shows the rail cart with RTL system on Track #2; Red arrows (Yellow outline) show directions for Test 1 (Point A to B) and Test 2 (Point B to A); Green dot on the track represents location of grease applicator.

Field test section 2 (Roanoke, VA)

The second round of low-speed field testing was carried out on an active revenue-service main line northeast of Roanoke, VA, in collaboration with Norfolk Southern. As shown in Figure 8, this double-track section was evaluated using Track #1. Unlike the Christiansburg site, the Roanoke segment features different track curvatures and a distinct grease formulation, allowing us to assess the RTL system's performance under varied field conditions. The test involved running the rail cart from Point A to Point B, along a 1.6-mile stretch with an upward elevation profile and a grease applicator positioned near the midpoint (marked by a green dot in Figure 8). After reaching Point B, the cart was turned around and driven back to Point A at speeds ranging from 2 to 4 mph. to measure the coefficient of friction (COF) on the flange face of the north rail. This was done to validate the performance of the RTL system by comparing its grease detection results with independent COF measurements, providing a reference for assessing detection accuracy. The results from the field tests, along with a comparison between the RTL system measurements and the tribometer readings, are presented and discussed in the following section.

Figure 8. Track layout of test section 2 (Roanoke, VA). Bottom left corner graph represents the speed of the cart along the distance travelled during the test; Bottom right image shows change in elevation between Point A and Point B recorded by GPS; Red arrows (Yellow outline) show directions for Test 1 (Point A to B) and Test 2 (Point B to A); Green dot on the track represents location of grease applicator; Image outlined in green shows the rail cart with RTL system on Track #1.

Field test results

This section presents the analysis of the detection results obtained from the low-speed field tests conducted in Christiansburg and Roanoke, VA. The focus is on evaluating the RTL system's ability to detect and differentiate grease levels under real-world conditions. To enhance the interpretation of the field test data, two color-coded graphs were developed to represent the lubricity conditions—unlubricated, lightly lubricated, and heavily lubricated—along both the north and south rails traversed by the cart. These visualizations were generated using the fluorescence intensity data recorded by the RTL system and provide a spatial overview of grease distribution across the test route. Additionally, rail curvature data, obtained from the GPS, was overlaid on these graphs to investigate potential correlations between track curvatures and lubrication effectiveness. This integrated visualization approach helps identify how curvature may influence grease application or retention on the rail surface.

Figure 9 presents the results from the field test conducted in Christiansburg, VA, with all plots displayed as a function of the total distance covered during the test. The top and bottom tricolor plots illustrate the lubrication conditions along the north and south rails of Track #2, respectively. These results correspond to the cart's traversal from west to east; therefore, the references to the north and south rails are made relative to the direction of travel. In these plots, red indicates unlubricated sections, yellow represents lightly lubricated areas, and green corresponds to heavily lubricated regions. The central plot represents the absolute degree of curvature along the same track segment. In the plots, the zero-foot mark corresponds to the location of the grease applicator. Negative distance values indicate the cart's approach toward the applicator, while positive values represent the cart moving away from it. The testing route consisted of 4 curves, with two left hand curves and two right hand curves. For right-hand curves, the north rail acted as the outer rail or the high rail of the curve—subject to greater lateral forces—while for the left-hand curve the south rail acted as the high rail. The results show that the high rail segments are predominantly represented in green, indicating heavy lubrication, while the low rail segments display more red and yellow regions, corresponding to unlubricated or lightly lubricated conditions. This pattern was consistent across all four curves encountered during the test. These findings suggest that greater amounts of grease were applied to the high rail compared to the low rail. Ensuring adequate lubrication on the high rail is essential to minimize rail wear, maintain proper wheel–rail interaction, and reduce the risk of damage or derailment.

Figure 9. Lubricity detection test results for test section 1 (Christiansburg, VA) in the form of tricolor plots. Top and bottom tricolor plots represent lubrication conditions for North and South rails respectively; Central plot shows absolute degree of curvatures of the track covered during the test; Green dotted line represents the location of grease applicator

To validate the grease-detection performance of the optical sensing system with various rail curvature and grease applicators, we collaborated with NS to conduct the second round of low-speed tests on the revenue-service track (Track #1) near Roanoke, VA shown in Figure 8. The NS track in Roanoke uses a different type of grease than the NS track in Christiansburg, allowing the system's performance to be evaluated across varying grease formulations.

Figure 10. Lubricity detection test results for test section 2 (Roanoke, VA) in the form of tricolor plots. Top and bottom tricolor plots represent lubrication conditions for North and South rails respectively; Central plot shows absolute degree of curvatures of the track covered during the test; Green dotted line represents the location of grease applicator

Figure 10 presents the results from the second field test, employing the same color-coded visualization approach used in Figure 9 for the first field test. In Figure 10 the results indicate that a limited amount of grease was found on the south rail. Grease was detected only on the high rail in Curve 2, with no detectable lubrication in the other curved sections. This outcome differs from the Christiansburg test, where grease was consistently identified on the flange face of the high rail across all curves, as shown in Figure 11. The absence of grease in the remaining curves of the second test is attributed to a malfunctioning applicator, which likely failed to dispense lubricant as intended.

Figure 11. Comparison of the lubricity detection results between the tests in Roanoke (Section 2) and Christiansburg (Section 1)

To verify the detection results, we reviewed footage captured by GoPro cameras mounted on the rail cart. A visual comparison of images from Test Sections 1 and 2, shown in the top two panels of Figure 12, revealed a clear contrast: fresh grease was visible near the applicator in Test Section 1 (Christiansburg), whereas the area surrounding the applicator in Test Section 2 (Roanoke) appeared dry. The middle images further support this observation—grease strings formed as the cart passed through the applicator in Test Section 1, indicating effective grease pickup by the wheel, a phenomenon not observed in Section 2. The bottom images in Figure 12 confirm these findings, showing visible grease along both the gauge corner and gauge face in Christiansburg, while the rail in Roanoke lacked proper lubrication. Both the sensing system and the camera indicate that the rail in Roanoke was not properly lubricated, and the applicator was not functioning properly. The NS division in Roanoke then conducted an examination on the applicator on Track #1 that was tested in Section 1. They found that the grease applicator indeed does not work due to the issue of the axle counter not being able to detect the presence of trains' axles. This further confirms the lubricity detection test results of the sensing system. Both the sensing system and camera footage consistently indicate that the rail in Roanoke was not adequately lubricated. Following this observation, the Norfolk Southern division in Roanoke inspected the grease applicator on Track #1 used in Test Section 2 and confirmed a malfunction. The issue was traced to a faulty axle counter that failed to detect train axles, preventing the applicator from dispensing grease. This validation further supports the accuracy and reliability of the RTL system in detecting insufficient lubrication.

Figure 12. Comparison of camera images from GoPro mounted on the rail cart, for assessing lubrication conditions between the tests in Roanoke (Section 2) and Christiansburg (Section 1)

Validation of system with tribometer measurements

During the testing conducted in Roanoke, VA (Test Section 2), a push tribometer from L.B. Foster was used by the vendor to measure the coefficient of friction (COF) on the flange face of the north rail. These measurements were taken in parallel with our system's data collection. This allowed for independent validation of the RTL system's grease-detection performance through direct comparison with tribometer readings. As shown in Figure 13, the tribometer data was collected in Curve #2 on the north rail. The results show strong alignment with the RTL system's detection output, as illustrated in the two plots on the right side of Figure 13. In areas identified as unlubricated by our system (marked in red), the tribometer recorded COF values well above 0.2, reaching up to 0.4—indicating essentially dry rail conditions. In lightly lubricated regions, COF dropped to around 0.15, and in heavily lubricated areas, it dropped below 0.15. This strong agreement demonstrates the reliability of the optical sensing system in detecting and distinguishing rail lubrication levels.

Figure 13. Comparison and validation of the sensing system's detection results with coefficient of friction (COF) measurement of tribometer for Section 2 (Roanoke, VA)

Conclusion

This study presents the results of field evaluation of a laser-based optical sensing system developed by RTL. The RTL system offers a reliable and scalable alternative to traditional grease assessment methods, which are often limited by subjectivity, limited coverage, and operational constraints. Laboratory testing confirmed the system's sensitivity to varying grease levels, with clear distinctions observed between lightly, moderately, and heavily lubricated sections. The results and comparison of two rounds of low-speed field tests conducted in collaboration with Norfolk Southern on Class I railroads in Christiansburg and Roanoke, VA are discussed and analyzed. These tests demonstrated the system's effectiveness in detecting grease presence under real-world conditions, including complex track geometries and variable grease application. Validation of the system's performance was supported through both visual and quantitative means. GoPro footage mounted on the rail cart provided clear visual confirmation of grease presence or absence, aligning well with the system's fluorescence-based detections. Additionally, tribometer readings taken in parallel showed strong correlation with the system's output, with friction levels matching the detected lubrication conditions.

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